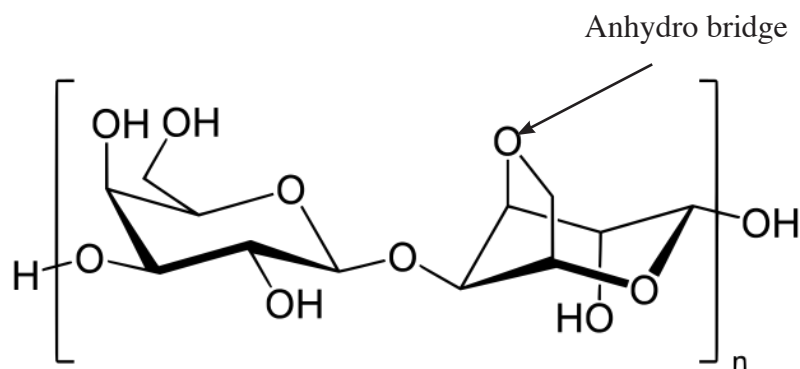
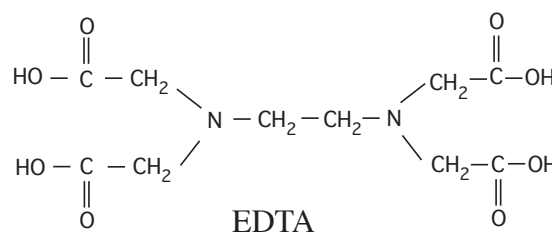


SOME MOLECULAR BIOLOGY REAGENTS

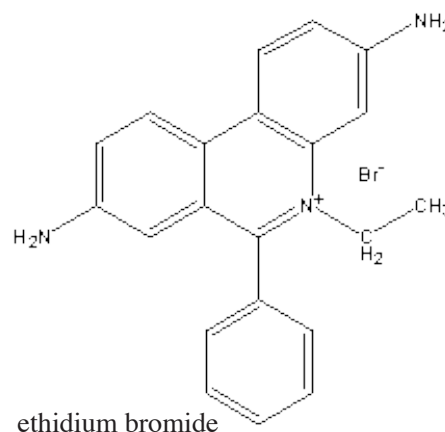
- agarose: is a polysaccharide, a linear helical polymer of hundreds of alternating D-galactose and L-galactose with frequent anhydro bridges between positions 3 and 6. It is extracted from various marine red algae (red seaweeds), typically *Gelidium* or *Gracilaria* species. Hydrogen bonding between polymers results in a 3D mesh (gel). Gel pore size is dependent on agarose concentration. The gel can be melted and solidified repeatedly by changing the temperature. An approximately 2% gel lightly flavored with sugar and almond is served as a dessert, Annindofu, in Japanese restaurants.



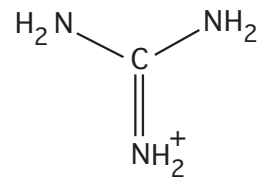
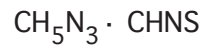
- ampicillin: A beta-lactam antibiotic closely related to penicillin. Like others in the group it kills sensitive bacteria by inhibiting the production of the bacterial cell walls. Resistant bacteria often produce an enzyme, beta-lactamase, that inactivates the antibiotic by cleavage.
- β -mercaptoethanol; $\text{CH}_2\text{SHCH}_2\text{OH}$: A reasonably strong reducing agent. It breaks cysteine disulfide bridges and so helps denature proteins. It stinks. Ethanethiol, a closely related compound, is added to natural gas as an odorant to make gas leaks detectable before they are dangerous.
- DNA gel loading buffer: 50% glycerol, xylene cyanol (light blue), Bromophenol blue (dark blue). This buffer increases the density of the sample and contains negatively charged dyes that migrate in the same direction as the DNA.
- DNase I: A powerful endonuclease that makes random cuts in dsDNA and ssDNA. It does not cut RNA at all.
- EDTA; ethylene diamine tetraacetic acid: is a chelating agent. It chelates (sequesters ie. prevents them from interacting with other things - kind of like putting them in jail.) divalent cations like Ca^{++} and Mg^{++} . Chelate comes from 'Chelos', the Greek word for crab. The acetic acid arms of EDTA are like crab legs that hold the cation.
- enzyme digestion buffer supplement (100 mM MgCl_2 , 50 mM NaCl)



- ethanol ($\text{CH}_3\text{CH}_2\text{OH}$): DNA and RNA are not soluble in alcohols, but water is. Both ethanol and isopropanol are therefore used to precipitate nucleic acids from aqueous solutions. It has other uses.
- ethidium bromide: A flat molecule (like a latke). It is a little bit hydrophobic and slips in between adjacent base pairs of double stranded DNA (or RNA). It is fluorescent and is used to stain DNA. It is certainly mutagenic and likely carcinogenic. If you want a laugh, look it up in the Merck index.
- Genomic DNA Extraction Buffer; (10 mM Tris-HCl, pH 8.0, 100 mM EDTA, 0.5% SDS, 0.5 mg/ml proteinase K, 20 $\mu\text{g}/\text{ml}$ RNase).

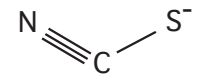


- guanidine isothiocyanate; guanidine hydrochloride: are both very strong denaturants (chaotropic agents) for proteins.



- isopropanol ($\text{CH}_3\text{CHOHCH}_3$): DNA and RNA are not soluble in alcohols, but water is. Both isopropanol and ethanol are therefore used to precipitate nucleic acids from aqueous solutions.

guanidine isothiocyanate



- lysozyme: An enzyme isolated from egg white. It lyses (breaks down) the bacterial cell wall. Its normal function is to prevent infection of the developing bird embryo or its food supply.
- Potassium Phosphate 2M (pH 7.4): A mixture of the monobasic and dibasic potassium salts of phosphoric acid. The sodium salt of the detergent SDS is soluble in aqueous solutions. The potassium salt is not.
- Protein gel loading buffer: (125 mM Tris (pH 6.8), 0.4% SDS, 10% β -mercaptoethanol, 20% glycerol, Bromophenol blue)
- Protein lysis buffer: (50 mM Tris-HCl, pH 8.0, 1 mM EDTA, 0.5% SDS)
- proteinase K: An enzyme that hydrolyzes proteins. It comes from a fungus, *Tritirachium album limber*. Unlike most enzymes, proteinase K retains its activity over wide pH and temperature ranges and is not disturbed by detergents or chaotropic agents. Related subtilisin proteases are added to laundry detergents.
- Resuspension buffer; (50 mM glucose, 10 mM EDTA, 25 mM Tris.HCl pH 8.0): This is just a buffer in which bacteria are reasonably happy and so do not lyse. The EDTA chelates divalent cations and prevents some types of protein-protein interactions that would otherwise cause the bacteria to stick to each other.
- Ribonuclease A - RNase: An enzyme which breaks down RNA. It is very, very difficult to inactivate.
- RNA binding membrane: These are all undefined. Why? This year, for the first time, the university has prevented us from letting you use a standard molecular biology reagent, phenol, in the student labs. It is somewhat dangerous and must be used carefully. So we now cannot teach you how to work with it and instead must rely on commercial kits that do not publish the composition of their reagents. In the kit you are using, the binding membrane is silica-based. Membrane surface silanol groups interact with the phosphates of DNA and RNA. This interaction is stronger if the salt concentration is high or in the presence of ethanol. In aqueous solutions, the hydrophobic silical surfaces and the hydrophobic surfaces of free bases can also interact. Proteins and carbohydrates do not bind to the membrane and can be washed through.
- RNA binding membrane desalting buffer:
- RNA binding membrane D`Nase inactivating buffer:
- RNA binding membrane wash buffer:
- RNA lysis buffer: Unknown contents.
- sodium chloride - NaCl: Salt, duh.
- sodium dodecylsulfate - SDS: A strong anionic detergent. It denatures proteins by interacting with the hydrophobic amino acid side chains that form the interior of most proteins and the peptide backbone. The protein becomes coated with the detergent. The net charge on the protein therefore becomes proportional to the number of SDS molecules bound.
- sodium hydroxide, NaOH: is a very strong chaotropic (Chaos - Greek, meaning disordered) agent. It completely denatures proteins, DNA and RNA by forming very strong hydrogen bonds that interrupt their normal hydrogen bondings.

- TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0): A common storage buffer for DNA. The tris buffer keeps the pH slightly alkaline. (Remember, DNA is an acid. If the pH is too low it will start to autohydrolyze.) The EDTA is often added to prevent the activity of any contaminating nucleases, most of which require either Mg^{++} or Ca^{++} for their activity.
- Tris, Tris-hydroxymethylaminomethane, $(HOCH_2)_3CNH_2$: is a commonly used biological buffer between pH 6.5 and 8.5. (It is not actually a very good buffer. We use it more for historical reasons than anything else.)
- Tris Borate EDTA running buffer; TBE: 54 g/l Tris, 27.5 g/l Boric Acid (0.89 M Tris-Borate), 0.1 mM EDTA.

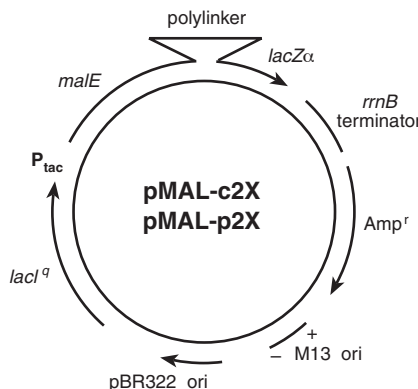
VECTORS:

MALTOSE BINDING PROTEIN (MBP) (FROM NEW ENGLAND BIOLABS)

The pMAL-c2 vector (Figure 1) provide a method for expressing and purifying a protein produced from a cloned gene or open reading frame. The cloned gene is inserted downstream from the *malE* gene of *E. coli*, which encodes maltose-binding protein (MBP), resulting in the expression of an MBP fusion protein. The *malE* sequence has been placed under the regulation of the the strong "tac" promoter and the *malE* translation initiation signals to give high-level expression of the cloned sequences. The expressed MBP fusion protein may easily be purified using MBP's affinity for maltose. The vectors express the *malE* gene (with or without its signal sequence) fused to the *lacZa* gene. Restriction sites between *malE* and *lacZa* are available for inserting the coding sequence of interest. Insertion inactivates the β -galactosidase a-fragment activity of the *malE-lacZa* fusion, which results in a blue to white color change on Xgal plates when the construction is transformed into an α -complementing host such as TB1 or JM107. When

Vectors

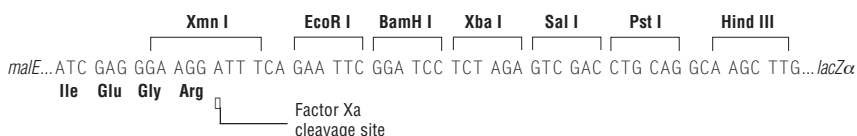
The pMAL-c2 series of vectors have an exact deletion of the *malE* signal sequence, resulting in cytoplasmic expression of the fusion protein. The pMAL-p2 series of vectors contain the normal *malE* signal sequence, which directs the fusion protein through the cytoplasmic membrane. All of the vectors include a sequence coding for the recognition site of a specific protease (Factor Xa [X], Enterokinase [E] or Genenase™ I [G]) which allows the protein of interest to be cleaved from MBP after purification.



pMAL™ Vectors

pMAL-c2X #N8076S	10
pMAL-p2X #N8077S	10
<i>Note: pMAL-c2X and pMAL-p2X are not suitable for purification systems.</i>	
pMAL-c2E (not sup) #N8066S	10
pMAL-p2E (not sup) #N8067S	10
pMAL-c2G (not sup) #N8068S	10
pMAL-p2G (not sup) #N8069S	10

pMAL™ -c2X, -p2X polylinker:



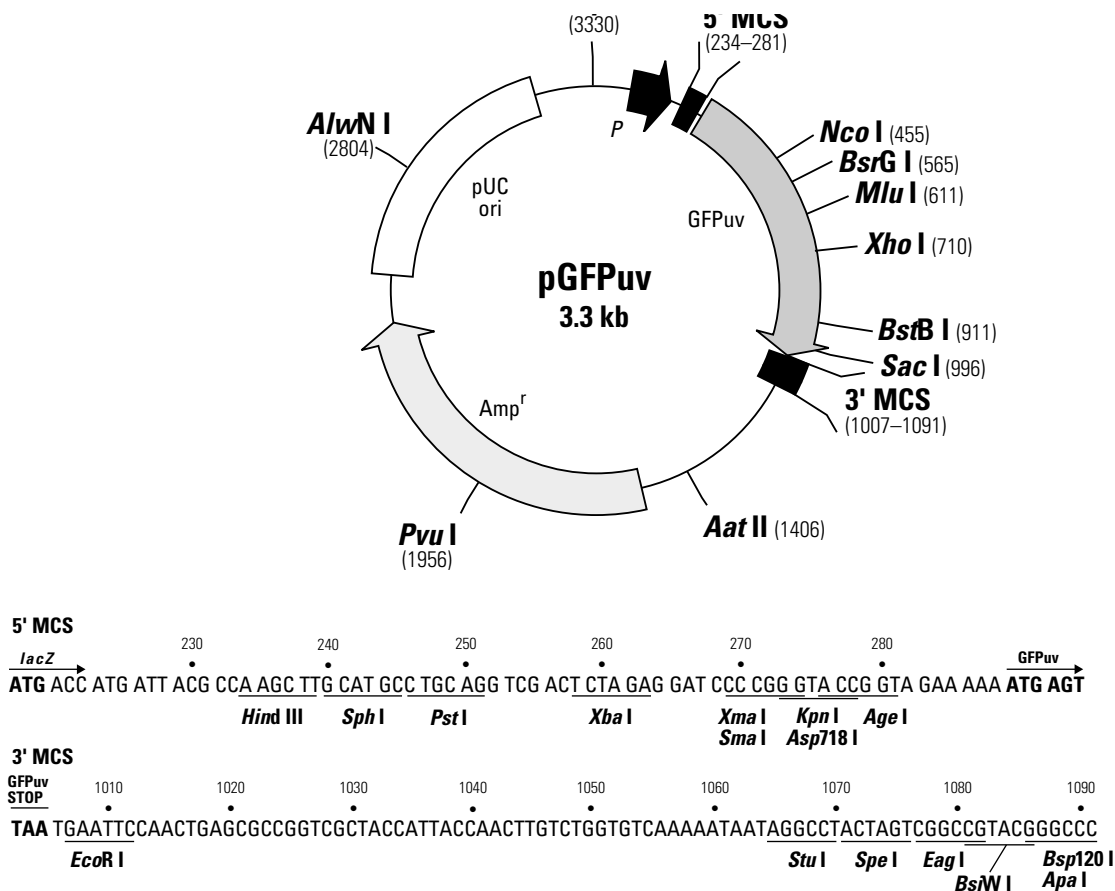
present, the signal peptide on pre-MBP directs fusion proteins to the periplasm. For fusion proteins that can be successfully exported, this allows folding and disulfide bond formation to take place in the periplasm of *E. coli*, as well as allowing purification of the protein from the periplasm. The pMAL™ -2 vectors also contain the sequence coding for the recognition site of the specific protease Factor Xa (9,10), located just 5' to the polylinker insertion sites. This allows MBP to be cleaved from the protein of interest after purification. Factor Xa cleaves after its four amino acid recognition sequence, so that few or no vector-derived residues are attached to the protein of interest, depending on the site used for cloning.

pGFPuv Vector Information

GenBank Accession #U62636

PT3055-5

Catalog #6079-1



Restriction Map and Multiple Cloning Site (MCS) of pGFPuv Vector. Unique restriction sites are in bold.

Description:

pGFPuv carries the "cycle 3" variant of GFP described by Cramer *et al.* (1). This gene was cloned between the two MCSs of the pUC19 derivative pPD16.43 (2). The GFPuv gene can be easily excised from pGFPuv. Alternatively, the GFPuv coding sequence can be amplified by PCR. The GFPuv gene was inserted in frame with the *lacZ* initiation codon from pUC19 so that a β -galactosidase-GFPuv fusion protein is expressed from the promoter in *E. coli*. Note, however, that if you excise the GFPuv coding sequence using a restriction site in the 5' MCS, the resulting fragment will encode the native (i.e., non-fusion) GFPuv protein. The pUC backbone of pGFPuv provides a high copy number origin of replication and ampicillin resistance gene for propagation in *E. coli*.

In the bioluminescent jellyfish *Aequorea victoria*, light is produced when energy is transferred from the Ca^{2+} -activated photoprotein aequorin to green fluorescent protein (GFP). The GFP chromophore consists of a cyclic tripeptide derived from Ser-Tyr-Gly in the primary protein sequence and is only fluorescent when embedded within the complete GFP protein. The crystal structures of GFP and GFP-S65T have revealed a tightly packed β -can enclosing an α -helix containing the chromophore. This structure provides the proper environment for the chromophore to fluoresce by excluding solvent and oxygen. Nascent GFP is not fluorescent, since chromophore formation occurs posttranslationally. The chromophore is formed by a cyclization reaction and an oxidation step that requires molecular oxygen. These steps are either autocatalytic or use factors that are ubiquitous, since fluorescent GFP forms in a broad range of organisms. Chromophore formation may be the rate-limiting step in generating the fluorescent protein, especially if oxygen is limiting. The wt GFP absorbs UV and blue light with a maximum peak of absorbance at 395 nm and a minor peak at 470 nm and emits green light maximally at 509 nm, with a shoulder at 540 nm.

The cloning of the wild-type GFP gene and its subsequent expression in heterologous systems has established GFP as a novel genetic reporter system. When expressed in either eukaryotic or prokaryotic cells and illuminated by blue or UV light, GFP yields a bright green fluorescence. Light-stimulated GFP fluorescence is species-independent and does not require any cofactors, substrates, or additional gene products from *A. victoria*. Additionally, detection of GFP and its variants can be performed with living tissues instead of fixed samples.

Many different GFP variant sequences have been developed to improve the fluorescence, the stability, reduce the toxicity and to localize the GFP protein to different cellular compartments. These variants are ideal for fluorescence microscopy and flow cytometry.

- Red-shifted GFP variants: EGFP - have red-shifted excitation spectra and fluoresce 4–35-fold more brightly than wt GFP when excited with blue light.
- EBFP: The blue fluorescent variant, EBFP, contain four amino acid substitutions that shift the emission from green to blue, enhance the brightness of the fluorescence, minimize rapid photobleaching, and improve solubility of the protein.
- GFPuv: When expressed in *E. coli*, GFPuv is 18 times brighter than wt GFP when excited by standard UV light and should be used in experiments using UV light for excitation. This variant contains additional amino acid mutations which also increase the translational efficiency of the protein in *E. coli*.